

Academic Schedule Automation Brief

A client-ready overview for automating schedule extraction, rendering, reminders, and email delivery from a live Excel workbook on Google Drive using Make.com.

Make.com architecture

Excel range rendering

Email reliability

Operational support

4

modular scenarios

2-5 days

estimated delivery

10-20 h

implementation range

0

silent failures tolerated

PROJECT SUMMARY

What the client needs automated.

The client maintains a live academic schedule in Excel on Google Drive and currently sends weekly, monthly, and tutor-specific schedule emails manually.

Current process

- Live Excel schedule stored on Google Drive.
- Manual weekly and monthly resident/tutor emails.
- Manual reminders based on teaching dates.
- High risk of missed updates and repetitive admin work.

Desired outcome

- Read the continuously updated workbook.
- Identify upcoming dates and relevant sections automatically.
- Preserve formatting exactly: colors, fonts, merged cells, layout.
- Send scheduled emails and reminders with minimal manual effort.

The real technical challenge

Sending email is not the hard part. The difficult part is preserving a heavily formatted, merged-cell academic schedule while keeping the workflow reliable over time.

The screenshot showed side-by-side groups, merged cells, section headers, color-coded blocks, tutor/location fields, and grouped week layouts. HTML email tables are risky because Outlook, Gmail, desktop, web, and mobile clients render complex tables differently.

Recommended framing for the call: this is an operations and rendering reliability project, not a simple email automation.

RECOMMENDED SYSTEM DESIGN

Make.com should orchestrate, not redraw the schedule.

The safest architecture is to use Make.com as the control layer while rendering the schedule from the spreadsheet source into stable image/PDF output.

1. Trigger

Scheduled weekly, monthly, reminder, or correction check.

2. Read

Workbook and helper/control sheet from Google Drive.

3. Select

Correct week, month, tutor session, or changed section.

4. Render

Excel range to image/PDF using the most reliable renderer.

5. Deliver

Email via Outlook/Gmail, log status, alert on failure.

Control layer

Make.com handles scheduling, branching, recipient resolution, deduplication, logging, retry paths, and admin alerts.

Rendering layer

Excel-compatible rendering generates a stable visual artifact so the schedule looks the same in Outlook, Gmail, and on mobile devices.

Data layer

A helper sheet or lightweight database stores week IDs, range references, tutor mappings, recipient groups, send status, and correction state.

Operational layer

Logs, unique send keys, rendered file links, source fingerprints, and failure alerts make the system maintainable by an administrator later.

Core position: prioritize formatting fidelity and operational clarity over clever HTML reconstruction. The schedule is the communication product.

PREFERRED STACK

Exact tools to test and why.

The final stack should be selected after testing against the real workbook, but the tool sequence should favor Excel-compatible rendering first.

LAYER	PREFERRED OPTION	REASON
Rendering	Microsoft Excel / OneDrive / Graph export, then PDF.co or CloudConvert	Best chance of preserving merged cells, colors, widths, fonts, and page layout without rebuilding the schedule manually.
Fallback render	Google Drive export only as fallback	Google Sheets can shift Excel formatting; useful if validated, risky if assumed.
Extraction	Anchor-based ranges plus helper/control sheet	Survives inserted rows better than hardcoded cell positions and keeps Make logic readable.
State tracking	Google Sheets, Airtable, Make Data Store, or Supabase for larger scope	Tracks what was rendered, sent, skipped, failed, or flagged for correction.
Corrections	Version/hash fingerprints per schedule section	Detects meaningful post-send changes and routes them to admin review first.

RECOMMENDED OUTPUT

Embedded image plus PDF attachment

The image makes the email easy to scan immediately. The PDF gives residents and tutors a reliable download/print version.

AVOID FIRST

Raw HTML table reconstruction

HTML can be flexible, but exact Excel-like rendering across Outlook desktop, Outlook web, Gmail, and mobile is unreliable for complex merged layouts.

MODULAR WORKFLOW PLAN

Split the system into focused scenarios.

Smaller Make scenarios reduce operational blast radius. If one workflow fails, the others can continue or fail clearly.

Weekly resident schedule

- Find next target week.
- Render the full weekly section.
- Send to resident groups.
- Write send log and rendered file link.

Monthly schedule

- Find upcoming month sections.
- Render month bundle as PDF.
- Send on configured schedule.
- Keep monthly send key separate from weekly keys.

Tutor reminders

- Resolve tutor sessions from mapping data.
- Send reminder based on teaching date.
- Use plain text details unless visual schedule is required.
- Alert if tutor name or email is unmapped.

Correction detection

- Compare schedule fingerprints after sends.
- Flag meaningful changes.
- Route to admin review before auto-correction.
- Prevent unnecessary correction emails for cosmetic edits.

Why modular scenarios matter

- Failures are easier to locate.
- Retry logic is simpler.
- Tutor reminders do not block weekly sends.
- Monthly rendering can fail without stopping corrections.
- Administrators can understand logs faster.
- Future changes can be isolated by scenario.

HANDLING WORKBOOK CHANGES

Use anchors and configuration, not fragile fixed ranges.

The workbook will evolve. The extraction design should expect inserted rows, wider schedules, additional resident groups, and new tutor logic.

ANCHOR STRATEGY

Identify sections by structure

- Week headers
- Date labels
- Block boundaries
- Group labels such as PGY1/PGY2
- Control-sheet range references where possible

CONTROL SHEET

Give Make clean data

- Week start and week end
- Range reference
- Resident group
- Tutor mapping
- Send/correction status

Weekly email	Extract and render the full relevant week section, from one structural marker to the next.
Monthly email	Bundle all upcoming month sections into a PDF attachment, optionally with preview images in the email body.
Tutor reminder	Resolve tutor sessions from structured rows/control sheet, then send only the relevant teaching date, time, topic, location, and group.
Correction	Compare current fingerprints against already-sent fingerprints and flag changes for admin review.

The extraction layer is the part most likely to require redesign if the workbook grows significantly. Rendering and email delivery usually adapt if the selected range is correct.

OPERATIONAL CONTROLS

Prevent duplicates, outdated reminders, and silent errors.

The system should prefer stopping and alerting an admin over silently sending the wrong schedule.

Duplicate prevention

Before each send, Make checks a unique key: email type, target week/month/session, recipient group, and schedule version. If the key exists, skip.

Outdated reminder prevention

Each reminder checks the latest workbook/control-sheet state before sending. Changed or cancelled sessions are skipped or flagged.

Rendering failure detection

After rendering, verify that the file exists, is not empty, and has the expected file type before any email goes out.

Admin visibility

Every run writes a log entry with source version, render link, recipient, send key, status, and error status.

STATE FIELD	PURPOSE
send_key	Prevents duplicate sends during reruns or repeated triggers.
source_version / fingerprint	Connects the sent email to the exact schedule version.
render_file_link	Allows admin review of what recipients received.
recipient_group / recipient_email	Makes delivery traceable and auditable.
status / error_message	Separates sent, skipped, failed, retrying, and needs-review states.

QUALITY ASSURANCE PLAN

Test rendering first, then delivery behavior.

Because formatting is the highest-risk part, the testing strategy should start with visual fidelity before recurring triggers are enabled.

Round 1: Render validation

- Render several different weekly sections.
- Compare against the Excel source.
- Check merged cells, colors, fonts, column widths, row heights, spacing, and page cropping.
- Confirm output is readable on mobile.

Round 2: Email client validation

- Outlook desktop
- Outlook web
- Gmail desktop
- Gmail mobile
- PDF download/print check

Go-live checklist

- Test mode completed with sample weeks.
- Recipient mapping table reviewed.
- Tutor names matched to email addresses.
- Duplicate send keys verified.
- Render files stored and accessible.
- Failure branches send admin alerts.
- Correction detection starts as review-only.
- Short admin documentation created.

Recommended go-live posture: run in test mode first, then enable recurring sends only after both visual comparison and email-client tests pass.

RISK REGISTER

Biggest risks and prevention plan.

Most long-term issues will come from operational changes: file moves, renamed columns, new row types, date format changes, or unmapped tutors.

RISK	PREVENTION
Formatting shifts during rendering	Use Excel-compatible renderer first; validate image/PDF output against real workbook before launch.
Wrong tutor reminders	Use explicit tutor mapping table, not free-form name inference from formatted cells.
Duplicate sends	Unique send keys and pre-send log lookup for every scenario.
Workbook edits break extraction	Anchor-based selection plus helper/control sheet; avoid fixed row numbers as the only logic.
Automated correction spam	Start with admin review instead of automatic correction emails.
Silent failures	Failure branches, admin alerts, retry handling, and explicit status logs.

Easiest to maintain

Recipient lists, email templates, scheduled triggers, tutor mappings, send logs, and reminder timing.

Hardest to maintain

Range detection, rendering fidelity, meaningful correction detection, inconsistent manual edits, and major workbook redesigns.

NEXT CONVERSATION

Plain-English talking points for the 30-minute call.

The client is evaluating long-term operational stability and maintainability. The call should demonstrate structured thinking, not overpromise perfect automation on day one.

Positioning statement

This workflow should be designed as a reliable operating system around the schedule: Make.com handles orchestration, a helper/control sheet provides clean logic, a renderer preserves visual fidelity, and a log/state layer prevents duplicate or incorrect sends.

Recommended answer on fidelity vs simplicity

For this use case, prioritize fidelity. If the schedule is visually wrong, the automation can create confusion even if it technically sends on time.

Recommended answer on corrections

Begin with admin review for changed schedules. Once the detection rules prove reliable, selected correction emails can be automated.

Recommended answer on future expansion

Wider schedules and new groups mostly affect extraction/configuration first. The rendering and email layers can usually stay stable.

Recommended answer on maintenance

The workflow should include visible logs, test mode, admin alerts, retry logic, unique send keys, and short documentation.

Meeting detail to confirm

Requested: Friday around 11:00 AM EST for a 30-minute call.

Suggested reply: confirm if available, or propose 2-3 alternative windows. Avoid vague availability if possible.